Challenge: Formatting Options

This document contains a new challenge for you using Power BI. This is not part of the ICA, but it is a self-assessment task to check that you are developing your reporting skills in Power BI. Once you have completed the tasks below, upload the Power BI project Padlet.

Below there is the set of tasks that you need to complete. Some tasks will require you to explore the Power BI tools or use Google to get to the final solution.
Try to complete this by yourself using the labs and lesson's resources covered in the past weeks, the solution will be made available to you on Blackboard.

This task uses data about food prices "wfp_market_food_prices". You can download the file from Blackboard. This file has been originally downloaded from Kaggle (https://www.kaggle.com/jboysen/global-food-prices)

1) Import the "food_prices" data
2) **Rename Some Columns**
   Use the information in the section "About this file" on Kaggle (https://www.kaggle.com/jboysen/global-food-prices) to **change the name of the columns**. You should identify and rename the following columns: **country, locality, month, year and price** (we are going to use just these columns, so don't worry about renaming the other columns).
3) Create a new column "Day" made of all ones (you can use the Custom Column tool in the Power Query Editor to do this – or Google "add a column of ones in power BI" to see if you can find a solution). If you're stuck, ask your tutor for help, or check the solutions at the end of this document.
   This will be for the first day of the month.

Day	month	year
1	1	2014
1	2	2014
1	3	2014
1	4	2014
1		
1		
1		
1		
1	9	2014
1	10	2014
1	11	2014
1	12	2014
1	1	2015
1	2	2015
1	3	2015

Are you stuck with this step? :)
Check my solution in the link below
Link to Solution

4) Click on Home-> Close and Apply to go close the Power Query Editor. Use the “year”, “month” and “day” columns to create **the calculated column** Date shown on the right. Use the DAX formula suggested here: https://community.powerbi.com/t5/Desktop/Month-Day-Year-Separate-Columns-Concatenate-to-Complete-Date/td-p/327594
When you use DAX, where do we do this? In Power Query Editor or in the main Power BI window?

Date
01/01/2015 00:00:00
01/05/2015 00:00:00
01/01/2015 00:00:00
01/02/2015 00:00:00
01/03/2015 00:00:00
01/04/2015 00:00:00
01/08/2015 00:00:00
01/09/2015 00:00:00
01/10/2015 00:00:00
01/05/2015 00:00:00
01/06/2015 00:00:00
01/10/2015 00:00:00
01/08/2015 00:00:00
01/04/2015 00:00:00
01/05/2015 00:00:00
01/06/2015 00:00:00

5) Now, move to report view.
Use country, locality and Date to create the **matrix** shown below.

Are you stuck with this step? :)
Check my solution in the link below
[Link to Solution]()

country	1992	1993	1994	1995
Afghanistan				
Algeria				
Armenia				
Azerbaijan				
Bangladesh				
Benin				
Bhutan				
Bolivia				
Burkina Faso				
Burundi				
Cambodia				
Cameroon				
Cape Verde				
Central African Republic				
Chad				
Colombia				
Congo				
Costa Rica				
Cote d'Ivoire				
Democratic Republic of the Congo				

6) **Drill down** on the columns to visualise the matrix below (only for 2016)
For a quick demo on how the Drill Up/Down Button works, you can check my video at the following link: [Link to video]()

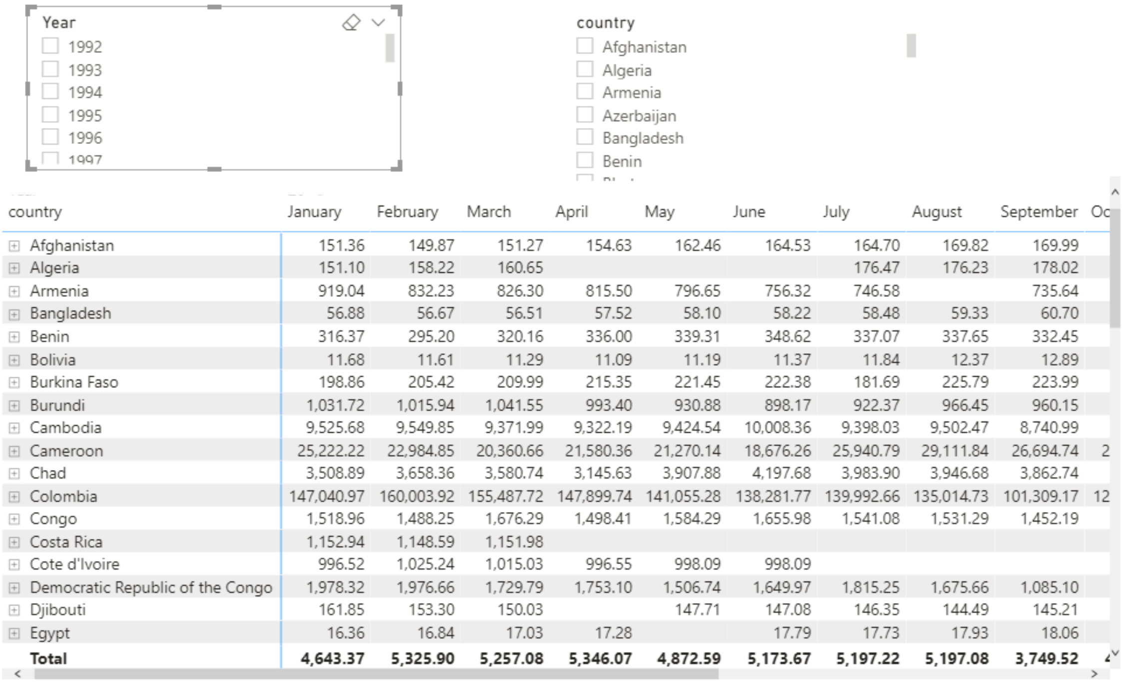
Drill on Columns

country	January	February	March	April	May	June	July	August	September	October	November	December
Afghanistan												
Algeria												
Armenia												
Bangladesh												
Benin												
Bolivia												
Burkina Faso												
Burundi												
Cambodia												
Cameroon												
Chad												
Colombia												
Congo												
Costa Rica												
Cote d'Ivoire												
Democratic Republic of the Congo												
Djibouti												
Egypt												
El Salvador												
Ethiopia												

7) Use the price column as values and show the **average** price

country	January	February	March	April	May	June	July	August	September	Oc
Afghanistan	151.36	149.87	151.27	154.63	162.46	164.53	164.70	169.82	169.99	
Algeria	151.10	158.22	160.65				176.47	176.23	178.02	
Armenia	919.04	832.23	826.30	815.50	796.65	756.32	746.58		735.64	
Bangladesh	56.88	56.67	56.51	57.52	58.10	58.22	58.48	59.33	60.70	
Benin	316.37	295.20	320.16	336.00	339.31	348.62	337.07	337.65	332.45	
Bolivia	11.68	11.61	11.29	11.09	11.19	11.37	11.84	12.37	12.89	
Burkina Faso	198.86	205.42	209.99	215.35	221.45	222.38	181.69	225.79	223.99	
Burundi	1,031.72	1,015.94	1,041.55	993.40	930.88	898.17	922.37	966.45	960.15	
Cambodia	9,525.68	9,549.85	9,371.99	9,322.19	9,424.54	10,008.36	9,398.03	9,502.47	8,740.99	
Cameroon	25,222.22	22,984.85	20,360.66	21,580.36	21,270.14	18,676.26	25,940.79	29,111.84	26,694.74	2
Chad	3,508.89	3,658.36	3,580.74	3,145.63	3,907.88	4,197.68	3,983.90	3,946.68	3,862.74	
Colombia	147,040.97	160,003.92	155,487.72	147,899.74	141,055.28	138,281.77	139,992.66	135,014.73	101,309.17	12
Congo	1,518.96	1,488.25	1,676.29	1,498.41	1,584.29	1,655.98	1,541.08	1,531.29	1,452.19	
Costa Rica	1,152.94	1,148.59	1,151.98							
Cote d'Ivoire	996.52	1,025.24	1,015.03	996.55	998.09	998.09				
Democratic Republic of the Congo	1,978.32	1,976.66	1,729.79	1,753.10	1,506.74	1,649.97	1,815.25	1,675.66	1,085.10	
Djibouti	161.85	153.30	150.03		147.71	147.08	146.35	144.49	145.21	
Egypt	16.36	16.84	17.03	17.28		17.79	17.73	17.93	18.06	
Total	**4,643.37**	**5,325.90**	**5,257.08**	**5,346.07**	**4,872.59**	**5,173.67**	**5,197.22**	**5,197.08**	**3,749.52**	**4**

8) Add a slicer for Year and a slicer for Country



country	January	February	March	April	May	June	July	August	September	Oc
Afghanistan	151.36	149.87	151.27	154.63	162.46	164.53	164.70	169.82	169.99	
Algeria	151.10	158.22	160.65				176.47	176.23	178.02	
Armenia	919.04	832.23	826.30	815.50	796.65	756.32	746.58		735.64	
Bangladesh	56.88	56.67	56.51	57.52	58.10	58.22	58.48	59.33	60.70	
Benin	316.37	295.20	320.16	336.00	339.31	348.62	337.07	337.65	332.45	
Bolivia	11.68	11.61	11.29	11.09	11.19	11.37	11.84	12.37	12.89	
Burkina Faso	198.86	205.42	209.99	215.35	221.45	222.38	181.69	225.79	223.99	
Burundi	1,031.72	1,015.94	1,041.55	993.40	930.88	898.17	922.37	966.45	960.15	
Cambodia	9,525.68	9,549.85	9,371.99	9,322.19	9,424.54	10,008.36	9,398.03	9,502.47	8,740.99	
Cameroon	25,222.22	22,984.85	20,360.66	21,580.36	21,270.14	18,676.26	25,940.79	29,111.84	26,694.74	2
Chad	3,508.89	3,658.36	3,580.74	3,145.63	3,907.88	4,197.68	3,983.90	3,946.68	3,862.74	
Colombia	147,040.97	160,003.92	155,487.72	147,899.74	141,055.28	138,281.77	139,992.66	135,014.73	101,309.17	12
Congo	1,518.96	1,488.25	1,676.29	1,498.41	1,584.29	1,655.98	1,541.08	1,531.29	1,452.19	
Costa Rica	1,152.94	1,148.59	1,151.98							
Cote d'Ivoire	996.52	1,025.24	1,015.03	996.55	998.09	998.09				
Democratic Republic of the Congo	1,978.32	1,976.66	1,729.79	1,753.10	1,506.74	1,649.97	1,815.25	1,675.66	1,085.10	
Djibouti	161.85	153.30	150.03		147.71	147.08	146.35	144.49	145.21	
Egypt	16.36	16.84	17.03	17.28		17.79	17.73	17.93	18.06	
Total	**4,643.37**	**5,325.90**	**5,257.08**	**5,346.07**	**4,872.59**	**5,173.67**	**5,197.22**	**5,197.08**	**3,749.52**	**4**

9) Change the slicers' format as dropdown, rather than list.

10)Use the slicers to select only 2016 and Countries: Armenia and Colombia.

Year

2016

country

Multiple selections

Drill on Rows

country	uary	February	March	April	May	June	July	August	September	October	Nove
Armenia	**919.04**	**832.23**	**826.30**	**815.50**	**796.65**	**756.32**	**746.58**		**735.64**	**769.00**	
	486.70	492.00	484.70	479.60	478.20	477.00	476.00		473.90	476.00	
Lori	924.74	842.09	829.81	805.35	789.46	751.26	748.44		718.72	741.86	
Tavush	912.52	827.60	826.38	816.48	799.58	765.68	754.13		742.09	800.49	
Yerevan	938.69	835.94	833.33	847.81	818.44	759.38			767.50	772.50	
Colombia	**7,040.97**	**160,003.92**	**155,487.72**	**147,899.74**	**141,055.28**	**138,281.77**	**139,992.66**	**135,014.73**	**101,309.17**	**122,384.69**	**122,**
Antioquia	7,180.55	133,712.68	138,354.81	130,011.13	130,141.23	121,935.10	124,802.42	118,470.81	106,513.45	103,052.45	107,
Cundinamarca	8,224.72	139,623.49	129,009.92	122,506.46	110,693.49	110,510.00	111,419.31	111,292.28	55,330.95	105,980.44	105,
Santander	2,013.56	249,440.89	242,362.94	233,726.67	225,635.61	226,606.56	228,062.56	214,905.67	191,965.72	191,221.67	186,
Total	**2,444.06**	**67,851.89**	**65,946.89**	**62,745.70**	**59,852.92**	**58,661.78**	**76,856.24**	**135,014.73**	**43,082.39**	**51,975.61**	**52,**

11) Under “Armenia” there are some missing values (blanks). Use the transform data tool to remove the blanks. The new table is shown below.

Drill on Rows

country	January	February	March	April	May	June	July	August	September	October	N
Armenia	**922.64**	**835.07**	**829.14**	**818.30**	**799.30**	**758.65**	**750.33**		**737.83**	**771.44**	
Lori	924.74	842.09	829.81	805.35	789.46	751.26	748.44		718.72	741.86	
Tavush	912.52	827.60	826.38	816.48	799.58	765.68	754.13		742.09	800.49	
Yerevan	938.69	835.94	833.33	847.81	818.44	759.38			767.50	772.50	
Colombia	**147,040.97**	**160,003.92**	**155,487.72**	**147,899.74**	**141,055.28**	**138,281.77**	**139,992.66**	**135,014.73**	**101,309.17**	**122,384.69**	**1**
Antioquia	127,180.55	133,712.68	138,354.81	130,011.13	130,141.23	121,935.10	124,802.42	118,470.81	106,513.45	103,052.45	
Cundinamarca	128,224.72	139,623.49	129,009.92	122,506.46	110,693.49	110,510.00	111,419.31	111,292.28	55,330.95	105,980.44	
Santander	222,013.56	249,440.89	242,362.94	233,726.67	225,635.61	226,606.56	228,062.56	214,905.67	191,965.72	191,221.67	
Total	**62,741.93**	**68,175.74**	**66,261.62**	**63,045.06**	**60,138.37**	**58,941.51**	**77,333.61**	**135,014.73**	**43,287.24**	**52,223.20**	

12) Change the selected countries to show only Afghanistan and Algeria

country	January	February	March	April	May	June	July	August	September	October	November	December	Total	Total
Afghanistan	**151.36**	**149.87**	**151.27**	**154.63**	**162.46**	**164.53**	**164.70**	**169.82**	**169.99**	**168.79**	**169.54**	**168.14**	**162.09**	**162.09**
$Daykundi	158.71	157.94	163.25	165.04	180.90	186.78	186.92	186.56	186.47	186.64	188.67	188.20	**178.01**	**178.01**
Badakhshan	157.80	150.16	151.25	175.24	181.45	174.74	175.34	174.76	175.24	165.34	152.13	141.33	**164.56**	**164.56**
Balkh	145.63	144.06	143.89	143.77	152.73	158.14	158.11	150.70	150.98	150.98	152.69	151.98	**150.30**	**150.30**
Faryab	159.74	158.62	158.59	158.64	174.58	179.84	180.10	179.97	180.19	180.12	180.56	181.21	**172.68**	**172.68**
Hirat	146.77	146.43	146.68	146.56	146.47	147.27	147.47	147.02	147.07	146.99	147.14	148.14	**147.00**	**147.00**
Kabul	154.06	153.91	159.22	160.64	176.69	182.15	182.47	181.97	181.91	182.28	183.73	183.33	**173.53**	**173.53**
Kandahar	142.37	142.49	142.08	141.81	141.63	142.17	141.95	192.65	193.07	192.97	193.49	192.99	**163.31**	**163.31**
Nangarhar	145.82	145.36	145.20	145.31	145.23	145.19	145.22	144.90	145.03	144.98	157.93	157.93	**147.34**	**147.34**
Algeria	**151.10**	**158.22**	**160.65**				**176.47**	**176.23**	**178.02**	**176.23**			**169.68**	**169.68**
Alger	142.76	127.90	133.43				142.81	146.52	145.43	141.00			**139.98**	**139.98**
Tindouf	153.64	163.76	165.63				181.63	180.79	183.01	181.63			**174.84**	**174.84**
Total	**151.20**	**155.79**	**157.92**	**154.63**	**162.46**	**164.53**	**173.39**	**174.55**	**175.92**	**174.28**	**169.54**	**168.14**	**166.62**	**166.62**

13) Create a DAX measure that displays the average price of all the countries. Use a card to display this value.

166.62

Average Prices

Are you stuck with this step? :)
Check my solution in the link below
[Link to Solution]

14) The next step is to change the colour of the cells based on their values (**using rules**). Any value greater than or equal to 166.62 should be green, any value less than 166.62 should be red (I used the #D0F0CO hex code for green, you can use a different one). Use this link if you need help changing the formatting **using rules** https://www.mssqltips.com/sqlservertip/6265/power-bi-conditional-formatting-for-matrix-and-table-visuals/

country	January	February	March	April	May	June	July	August	September	October	November	December	Total	Total
Afghanistan	**151.36**	**149.87**	**151.27**	**154.63**	**162.46**	**164.53**	**164.70**	**169.82**	**169.99**	**168.79**	**169.54**	**168.14**	**162.09**	**162.09**
$Daykundi	158.71	157.94	163.25	165.04	180.90	186.78	186.92	186.56	186.47	186.64	188.67	188.20	**178.01**	**178.01**
Badakhshan	157.80	150.16	151.25	175.24	181.45	174.74	175.34	174.76	175.24	165.34	152.13	141.33	**164.56**	**164.56**
Balkh	145.63	144.06	143.89	143.77	152.73	158.14	158.11	150.70	150.98	150.98	152.69	151.98	**150.30**	**150.30**
Faryab	159.74	158.62	158.59	158.64	174.58	179.84	180.10	179.97	180.19	180.12	180.56	181.21	**172.68**	**172.68**
Hirat	146.77	146.43	146.68	146.56	146.47	147.27	147.47	147.02	147.07	146.99	147.14	148.14	**147.00**	**147.00**
Kabul	154.06	153.91	159.22	160.64	176.69	182.15	182.47	181.97	181.91	182.28	183.73	183.33	**173.53**	**173.53**
Kandahar	142.37	142.49	142.08	141.81	141.63	142.17	141.95	192.65	193.07	192.97	193.49	192.99	**163.31**	**163.31**
Nangarhar	145.82	145.36	145.20	145.31	145.23	145.19	145.22	144.90	145.03	144.98	157.93	157.93	**147.34**	**147.34**
Algeria	**151.10**	**158.22**	**160.65**				**176.47**	**176.23**	**178.02**	**176.23**			**169.68**	**169.68**
Alger	142.76	127.90	133.43				142.81	146.52	145.43	141.00			**139.98**	**139.98**
Tindouf	153.64	163.76	165.63				181.63	180.79	183.01	181.63			**174.84**	**174.84**
Total	**151.20**	**155.79**	**157.92**	**154.63**	**162.46**	**164.53**	**173.39**	**174.55**	**175.92**	**174.28**	**169.54**	**168.14**	**166.62**	**166.62**

15) The final step is to change the formatting of the slicers adding a text box as shown below.

Price Development

Year

2016

country

Multiple selections

country	January	February	March	April	May	June	July	August	September	October	November	December	Total	Total
Afghanistan	**151.36**	**149.87**	**151.27**	**154.63**	**162.46**	**164.53**	**164.70**	**169.82**	**169.99**	**168.79**	**169.54**	**168.14**	**162.09**	**162.09**
$Daykundi	158.71	157.94	163.25	165.04	180.90	186.78	186.92	186.56	186.47	186.64	188.67	188.20	**178.01**	**178.01**
Badakhshan	157.80	150.16	151.25	175.24	181.45	174.74	175.34	174.76	175.24	165.34	152.13	141.33	**164.56**	**164.56**
Balkh	145.63	144.06	143.89	143.77	152.73	158.14	158.11	150.70	150.98	150.98	152.69	151.98	**150.30**	**150.30**
Faryab	159.74	158.62	158.59	158.64	174.58	179.84	180.10	179.97	180.19	180.12	180.56	181.21	**172.68**	**172.68**
Hirat	146.77	146.43	146.68	146.56	146.47	147.27	147.47	147.02	147.07	146.99	147.14	148.14	**147.00**	**147.00**
Kabul	154.06	153.91	159.22	160.64	176.69	182.15	182.47	181.97	181.91	182.28	183.73	183.33	**173.53**	**173.53**
Kandahar	142.37	142.49	142.08	141.81	141.63	142.17	141.95	192.65	193.07	192.97	193.49	192.99	**163.31**	**163.31**
Nangarhar	145.82	145.36	145.20	145.31	145.23	145.19	145.22	144.90	145.03	144.98	157.93	157.93	**147.34**	**147.34**
Algeria	**151.10**	**158.22**	**160.65**				**176.47**	**176.23**	**178.02**	**176.23**			**169.68**	**169.68**
Alger	142.76	127.90	133.43				142.81	146.52	145.43	141.00			**139.98**	**139.98**
Tindouf	153.64	163.76	165.63				181.63	180.79	183.01	181.63			**174.84**	**174.84**
Total	**151.20**	**155.79**	**157.92**	**154.63**	**162.46**	**164.53**	**173.39**	**174.55**	**175.92**	**174.28**	**169.54**	**168.14**	**166.62**	**166.62**

Here is another link that you might find useful when working with cells formatting:
https://docs.microsoft.com/en-us/power-bi/create-reports/desktop-conditional-table-formatting

Once you have completed this task, add any other charts that you think might improve your dashboard and share it on Padlet.

Consider using the skills learnt in this challenge for your ICA.

Solution Video

Did you get stuck in any of the steps above? Check the solution below.
Before clicking the link…are you sure you don't want to have another go at it before checking the answer? :)

Link to Solution